

Mia's moving day · Student handouts

Session 1 · Sustainable Supply Chain Management · TH Köln

Handout 1: Mia's moving day

Mia studies engineering. On Saturday she moves out of her flat on the third floor. There is no lift. Her new flat is on the ground floor, 15 km away.

She has booked a rental van from 09:00. At 18:00 the van must be back at the rental depot, next to the old flat, and the keys must go back to the landlord. The landlord will check that the old flat is clean. If Mia is late, the rental firm charges 30 € for every started hour.

Two friends, Jonas and Sara, have offered to help for pizza and drinks. More friends could come too. Mia has saved 120 € for the whole day. It is exam time, so she has only 4 free hours in the week before to pack boxes.

Mia does not know how to plan this. A small van is cheap but has to drive twice. A large van drives once but costs more. Every friend who helps is faster, but costs money. Packing early saves time on Saturday, but Mia does not have the hours to pack everything.

Your job: plan Mia's day as a team. Choose a van, a crew and a packing plan. Show with numbers why your plan works. There is no single right answer. There are better and worse answers, and you must be able to explain yours.

Handout 2: Data sheet

How the day works

1. The clock starts at 09:00. The crew works on one task at a time, in the order of the table below.
2. Work is measured in person-minutes. 360 person-minutes is one person for 6 hours, or three people for 2 hours.
3. Time for a task = work ÷ number of people working. Example: carrying down is 270 person-minutes. With 3 people it takes $270 \div 3 = 90$ minutes.
4. Each task has a maximum number of people. More people than the maximum do not help. Example: with 5 people only 4 can carry, so carrying takes $270 \div 4 = 67.5$ minutes.
5. One person drives the van. One drive takes 15 minutes. A trip is old flat, new flat, and back to the old flat: 30 km.
6. After the last unloading, with 3 or more people: one person drives the van back and cleans the old flat alone. The others reassemble the furniture. With 2 people: both reassemble first, then drive back and clean together.
7. The whole flat is 100 moving units (boxes and furniture pieces). That is the output of the day.

The tasks

Order	Task	Work (person-minutes)	Max people	Note
1	Pack boxes	360	3	Mia can pack part of it in the week before
2	Dismantle bed and shelves	90	2	
3	Carry down, three floors	270	4	Split equally over the trips
4	Load the van	120	3	Split equally over the trips
5	Drive, one way	15 minutes	1	Then unload; if a second trip is needed, drive back first
6	Unload at the new flat	90	3	Split equally over the trips
7	Reassemble furniture	135	3	At the new flat
8	Clean the old flat	120	2	Only after the last load. One person alone needs 120 minutes

Your choices and what they cost

Choice	Options	Cost
Van	Small: holds half the flat, so 2 trips (60 km). Large: holds everything, 1 trip (30 km)	Small: 45 € per day + 0.30 € per km. Large: 75 € per day + 0.40 € per km
Crew	Mia plus 1 to 4 friends (crew of 2 to 5 people)	15 € per friend for pizza and drinks. Mia is free
Packing in the week before	Nothing, half the boxes (3 hours of Mia's time), or all boxes (6 hours)	No cash cost. Mia has only 4 free hours, so "all boxes" is not possible
Return slot	Until 18:00, or until 20:00	18:00 is included. 20:00 costs +20 €
Late fee	After the return slot	30 € for every started hour

Budget: 120 € for van, friends and return slot. The late fee is not in the plan, but it is money lost if the day goes wrong.

Handout 3: Worksheet, two plans by hand

Calculate both plans with a calculator. Write the clock time after every task.

Plan A: small van, crew of 3 (Mia and two friends), nothing packed in the week before.

Plan B: large van, crew of 3 (Mia and two friends), half of the boxes packed in the week before (Mia spends 3 hours).

Step 1. Schedule. Use the order and the rules of Handout 2. Start at 09:00. Write down when the last unloading ends, when the old flat is clean, and when the furniture is reassembled. The finish time is the later of the two.

Step 2. Cost. Van day price + kilometres $\times$ price per km + 15 € for each friend.

Step 3. Productivity. The output is 100 moving units in both plans. Productivity is output divided by input, so each measure below has a different input.

Measure	How to calculate	Plan A	Plan B
Finish time	Step 1		
Minutes to spare before 18:00	18:00 minus finish time		
Cost (€)	Step 2		
Person-hours on Saturday	crew $\times$ hours from 09:00 to finish		
Units per person-hour, Saturday	$100 \div$ person-hours on Saturday		
Units per person-hour, whole project	$100 \div$ (person-hours on Saturday + Mia's packing hours in the week before)		
Units per euro	$100 \div$ cost		

Questions

1. Which plan is more productive on Saturday? Which is more productive for the whole project?
2. Which plan is more productive per euro? Why can the answers to questions 1 and 2 disagree?
3. Both plans fit the budget. Which would you choose, and what are you giving up?
4. Mia's friend Sara says: "Let's ask two more friends, then we finish sooner." Is she right? Show it with numbers.

Bonus. Crew utilisation = work content $\div$ person-hours on Saturday. Work content is the sum of all work on Saturday (from the task table, minus what was packed early) plus the driving time of the driver. Which plan keeps its crew busier?

Handout 4 (optional, for quick teams): When things go wrong

A plan is one thing. The real day is another. Every team draws event cards and applies them to its own plan, by hand or in the Colab model. Cut the table into cards.

Card	Event	Effect on the plan
1	Blocked stairwell. A neighbour's delivery blocks the stairs.	Carrying down takes 70 % longer
2	A friend cancels. He wakes up ill on Saturday morning.	The crew is one person smaller for the whole day
3	Late van. The rental firm delivers the van to the door 60 minutes late.	The van is not there at 09:00. What can the crew do meanwhile?
4	Stuck bed frame. The screws will not turn.	Dismantling takes 30 minutes longer
5	Rain. Boxes and furniture must be covered and carried slowly.	Carrying down and loading take 30 % longer
6	Failed inspection. The landlord sends Mia back to clean the oven again.	Cleaning takes 45 minutes longer, done by one person

For each card, answer: Does your plan still finish by the end of the return slot? If not, what does the late fee cost? Which task is now on the critical path, that is, the task that decides the finish time?

Then decide where to put a buffer. You can buy time (return slot until 20:00, +20 €), buy capacity (one more friend, +15 €, if the budget allows), or move work out of the day (pack more in the week before, if Mia has hours left). You pay with one of the three.

Finish this sentence in your notebook or on paper: "Our plan finishes at ____ on a normal day. On about ____ days out of 10 we are late, so we ____."

AI check. If you used an AI assistant today, write one sentence: what did you check in its answer, and what did you change?

Source: Eigene Darstellung, R. Bartnik (2026). Definitions of productivity: Heizer, J., Render, B., & Munson, C. (2023). *Operations management: Sustainability and supply chain management* (14th ed., Global ed.). Pearson, Chapter 1.